

A SHIELDED BOX FOR THE DIPOLE DETECTOR

Plain-English build guide for Steve · full details and sources in CAGE_ESTIMATE.md

THE ONE RULE: NOTHING GOES THROUGH THE WALL.

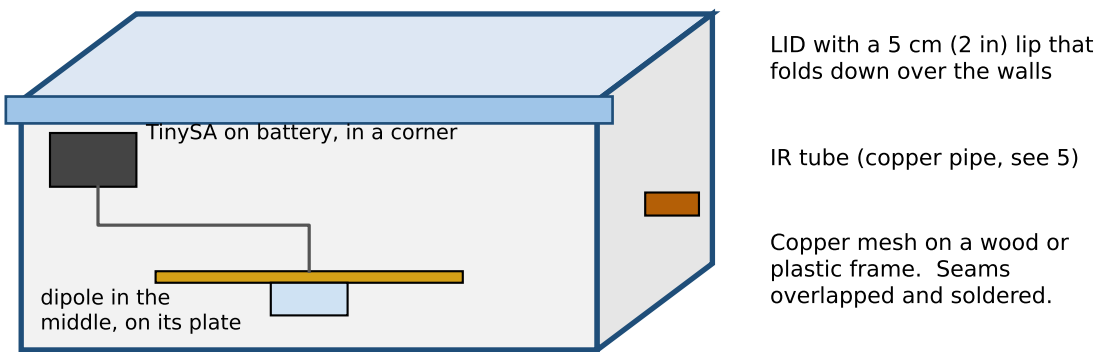
No cables, no connectors, no feed-throughs. The TinySA runs on its own battery inside; save the reading and read it after opening the lid. The box keeps ordinary radio out; the dipole is its own instrument with its own box (not shared with the coil cage, the Experiment B rods, or the oven).

1. WHICH SIZE (inside dimensions)

Box	Use for	Gap to walls	OK?
60 × 60 × 60 cm (24 × 24 × 24 in)	DD-1296, DD-2450	≥ 1.06 wavelength	YES — Option 1
100 × 60 × 60 cm (39 × 24 × 24 in)	DD-433 (and the other two)	≈ 0.43–0.49 wavelength	YES — Option 2
30 cm cube, or 40 × 20 × 15 cm	—	far too close	NO
120 × 60 × 60 cm	—	box itself rings at 433 MHz	AVOID

Why: metal walls closer than about half a wavelength to one wavelength change how the dipole tunes (rule of thumb). Even in the right box, re-check the dipole's tuning *INSIDE* the closed box and write it down.

2. WHAT IT LOOKS LIKE



3. MATERIAL (pick one)

- BEST: 16-mesh copper screen (TWP). You can solder the seams.
- FASTEST: nickel-copper shielding fabric (Less EMF), sewn or taped into a bag with a roll-down top.
- CHEAPEST: aluminium window screen (\$41.55 a roll). Can't be soldered, so seams are weaker.
- How much: Option 1 about 3.2 m² (34 ft²), Option 2 about 4.5 m² (48 ft²), incl. overlaps + 20 % spare.

4. BUILD TIPS

- Overlap every seam 2–5 cm and solder it (copper), or cover both sides with copper tape that has CONDUCTIVE glue.
- The lid must touch all the way round — not just at a few clamps. Use metal finger strip or a conductive gasket, or press the lip flat with a wooden strip clamped at each end. Any hole or gap: keep it under 6 mm.
- No ground wire needed: a closed box with no holes shields without one. A ground on the OUTSIDE is optional.

5. IR THERMOMETER HOLE

3/4 in Type L copper pipe (inside 20 mm), 80 mm (3.15 in) long, soldered all round into a copper plate in the wall. It blocks ordinary radio (about 128 dB by the standard formula) but light and infrared go straight through. Put the IR gun right against the pipe. Keep a copper end cap: readings open vs capped are also a test.

COST + TIME (estimate)

Option 1: about \$155–310, 9–12 hours
Option 2: about \$190–420, 12–16 hours (fabric bag: 3–6 h)

Dipole Detector cage · Rev A 2026-09-27

Grok Bot initial estimate — VERIFY
Prepared for Dan Britton (builder: Steve)